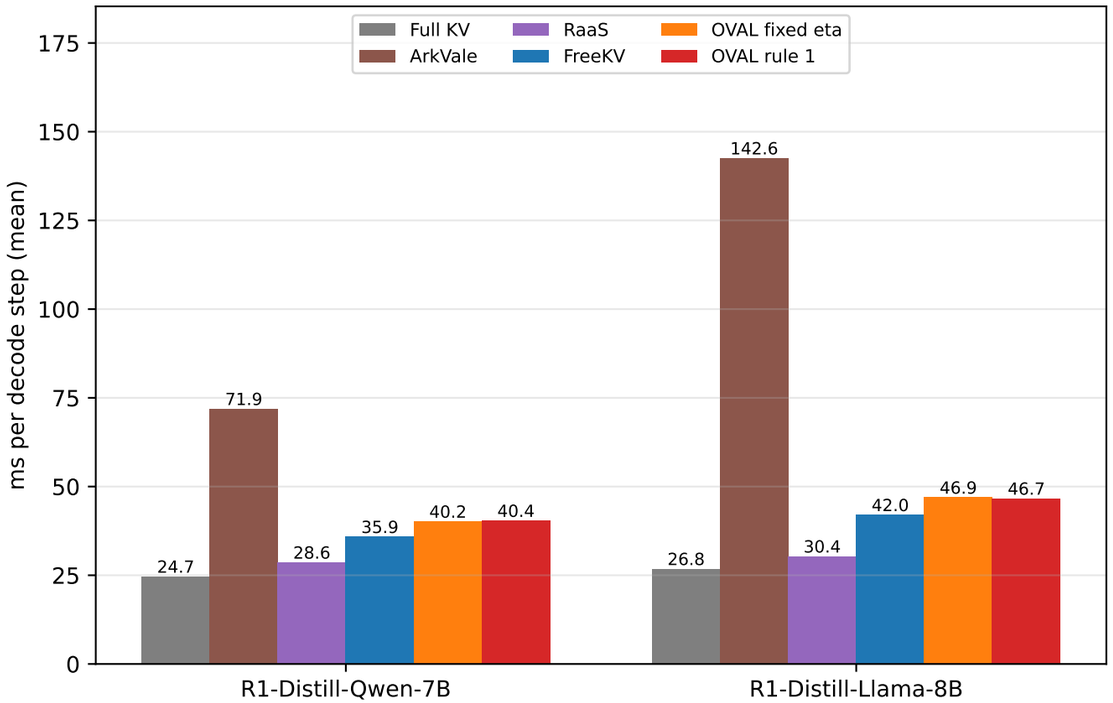
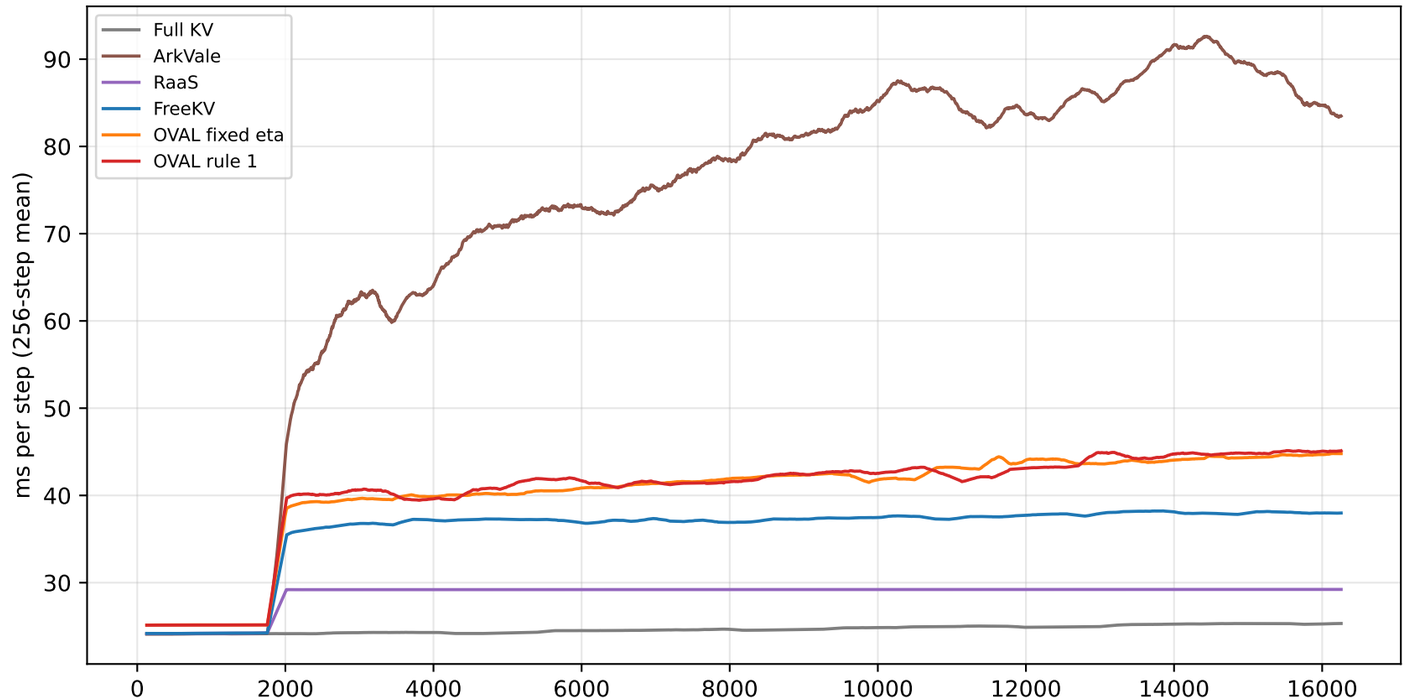
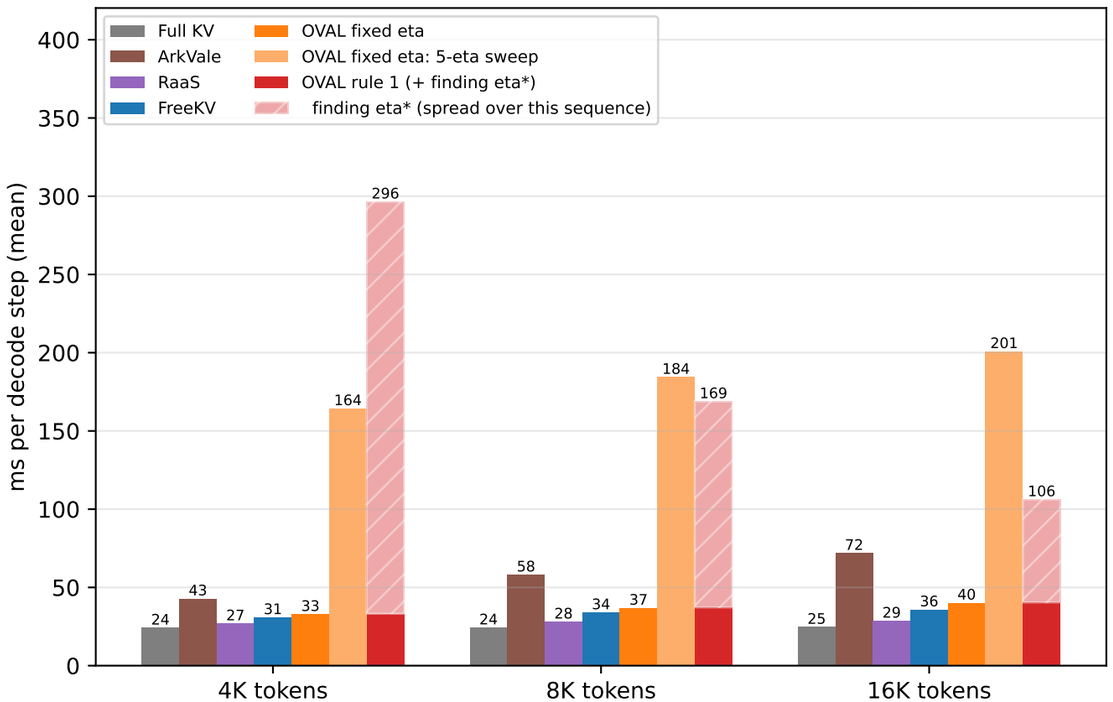
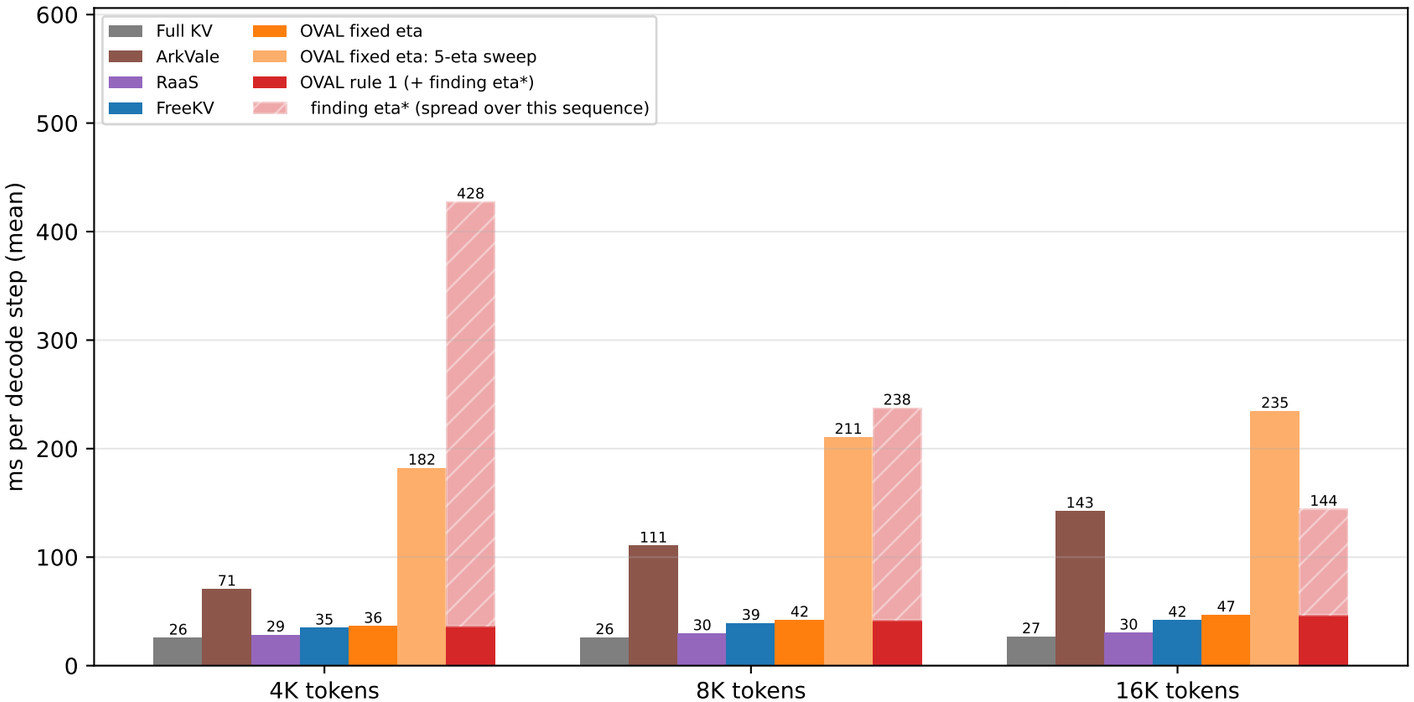

Decode latency, AIME24 problem 0, 16384 decoded tokens, batch 1
budget 2048 tokens (incl. 512 sink + 512 recent), A6000
Full KV
ArkVale
RaaS
FreeKV
OVAL fixed eta
OVAL rule 1
ms per decode step (mean)
0
25
50
75
100
125
150
175
24.7
71.9
28.6
35.9
40.2
40.4
26.8
142.6
30.4
42.0
46.9
46.7
R1-Distill-Qwen-7B
R1-Distill-Llama-8B


R1-Distill-Qwen-7B: per-step decode time over the generation
Full KV
ArkVale
RaaS
FreeKV
OVAL fixed eta
OVAL rule 1
ms per step (256-step mean)
30
40
50
60
70
80
90
0
2000
4000
6000
8000
10000
12000
14000
16000
decode step


R1-Distill-Qwen-7B: decode time by sequence length
sweep = 5 decodes (one per eta); rule 1 = 1 decode + finding eta*
Full KV
ArkVale
RaaS
FreeKV
OVAL fixed eta
OVAL fixed eta: 5-eta sweep
OVAL rule 1 (+ finding eta*)
finding eta* (spread over this sequence)
ms per decode step (mean)
0
50
100
150
200
250
300
350
400
24
43
27
31
33
164
296
24
58
28
34
37
184
169
25
72
29
36
40
201
106
4K tokens
8K tokens
16K tokens


R1-Distill-Llama-8B: decode time by sequence length
sweep = 5 decodes (one per eta); rule 1 = 1 decode + finding eta*
Full KV
ArkVale
RaaS
FreeKV
OVAL fixed eta
OVAL fixed eta: 5-eta sweep
OVAL rule 1 (+ finding eta*)
finding eta* (spread over this sequence)
ms per decode step (mean)
0
100
200
300
400
500
600
26
71
29
35
36
182
428
26
111
30
39
42
211
238
27
143
30
42
47
235
144
4K tokens
8K tokens
16K tokens